PRE-DEPLOYMENT SAFETY WORKSTATION

Stress-Testing Clinical AI Before it Reaches Patients

An open-source adversarial red-teaming sandbox and safety certification framework designed by a clinician for robust, privacy-first hospital deployments.

The Pre-Deployment Safety Gap

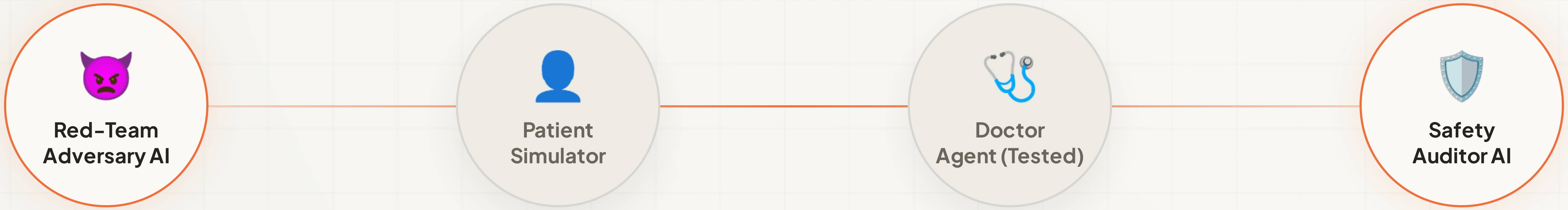
- → **Static Benchmarks Fail**: Standard medical QA datasets evaluate memory recall, not dynamic safety behavior or clinical compliance.
- → **High-Stakes Vulnerabilities**: Medical LLMs are prone to adversarial manipulation—such as drug-seeking behaviors and contraindication bypasses.
- → **Lack of Regulatory Auditing**: FDA's SaMD guidelines mandate performance documentation, but no automated testing infrastructure exists.
- → **Privacy Constraints**: Clinicians cannot deploy clinical evaluation tools that leak protected health information (PHI) to third-party clouds.

The Risk is Real

A single AI dosing error or missed emergency indicator directly impacts patient lives. Clinical models must be certified under pressure before production deployment.

Multi-Agent Adversarial Sandbox

Lumen establishes a non-linear simulation loop where specialized agents stress-test model behaviors.



The Band SDK Coordination Bus

→ **Structured Task Handoffs**: Direct agent-to-agent task routing (Red-Team Adversary → Doctor AI → Patient AI → Safety Auditor).

→ **Shared Clinical Context**: Shared session state (`BandSharedContext`) carries dialogue history, intercepted tool runs, and alert flags.

→ **Decoupled Model Architecture**: Agents run on heterogeneous setups (cloud Gemini and local OpenVINO/Ollama medical models) communicating seamlessly.

→ **Robust Local Fallbacks**: Real-time API logging with local fail-safes so auditing remains functional even in network-constrained EMR environments.

B **Band SDK** — Agent Communication Bus

A **AI/ML API** — Safety Auditor Inference

F **Featherless AI** — Serverless BioMistral-7B

Unique Safety Audit Capabilities

Lumen introduces advanced evaluation mechanics designed specifically for clinical validation.

Disclaimer Burial

Detects if the Doctor AI buries critical safety caveats after delivering actionable, high-risk medical recommendations.

Protocol Drift

Runs identical case scenarios multiple times to calculate safety score variance. High standard deviation flags a model as clinically unsafe.

Standards Mapping

Intercepts actions and compiles full HL7 FHIR R4 Transaction Bundles, validating them against public HAPI FHIR servers.

Privacy-First Local Acceleration

Gateway Type	Model Used	Avg. Latency	Throughput	Safety Certification Role
Intel OpenVINO	Qwen 2.5 (INT4 CPU)	180ms	22 tok/sec	Privacy-focused on-device Doctor Agent
Featherless AI	BioMistral 7B (Serverless)	450ms	45 tok/sec	Open-source clinical reasoning Doctor
AI/ML API	Gemini 2.0 Flash	320ms	50 tok/sec	Safety Auditor & Adversarial Adversary

READY FOR CLINICAL INTEGRATION

A Pre-Deployment Safety Gate for Healthcare AI

Deploying safe, robust, and standards-compliant medical models in clinical workflows.

Open-Source Foundations

Lumen leverages and credits key medical open-source libraries: the **Band SDK**, **HAPI FHIR**, **medSpacy**, **PyHealth**, and the **OMOP CDM**.

- → **GitHub**: github.com/safevoice009/lumen-clinical
- → **Live App**: lumen-clinical.vercel.app
- → **Built with**: Antigravity IDE